

How to not get lost when papers browsing – a bibliography guide



Why are you doing bibliography?

- Depth: **Understand key SOTA concepts**
 - Content: Reference high quality papers
 - Method: in depth reading → 🕒⭐
- Breadth: **Build a mental picture of a field you don't know**
 - Content: Lots of varied quality papers starting from the surveys
 - Method: skimming/linking → ⚡📖
- Try to **fix a problem**
 - Content: Key word related/referenced papers/asking for references
 - Method: selective reading/comparison tables → 👁👁
- Try to **find new problems!** (Same as breadth)
- **Keep up with news**
 - Content: Twitter/colleagues suggestions
 - Method: depends on whether it's curiosity based or field related

How to do bibliography?

- How to look for papers?
- How to read a paper?
- How to store information about a paper read?

How to look for papers

- Basics:
 - Google scholar
 - Semantic scholar
 - Connected papers
 - **Citations in good papers!**
 - **Colleagues!**
 - **Surveys!**
- News:
 - Social networks: twitter
 - Following arxiv with rss feeds
 - Setting up google scholar alerts
- Imo, **avoid AI based paper summarization search engines** (ex: consensus/elicit/...)

How to read a paper

- **Skimming**
 - Read: Intro/Abstract/Table of contents/Figures/Equations/Conclusion
 - Identify paper type/context/validity/contributions → 💎/💩
- Basic reading (**think like a student**)
 - Understand the paper and concepts
 - Store references
- In depth reading (**think like a reviewer**)
 - Criticize
 - “Re-write”, learn about the story telling
 - Link to what you know, cross check
- All the time, take notes!

Always take notes!

**When You've Forgotten
Something...**



...But Can't Remember What.

Why we want to store information

- Goals:
 - “Reload” the paper fast in your memory/avoid information loss
 - **Paper notes**, quotes, ...
 - Create your mental map of the domain/connect
 - **Table of contents**
 - **Tags**/connections
 - Do bibliography efficiently
 - **Quality metadata**
 - Summaries

How to store information

- A markdown tool for your notes with **backlinks** and a **tagging system**
 - ex: Obsidian, Zettlr
 - Or Sublime Text + Zotero ^^”
- **+ a ToC page somewhere**
- **A metadata manager**
 - **Zotero + zotero connector + better bib tex**

1. Reload: Examples of notes

Sprague et al_2024_MuSR Testing the Limits of Chain-of-thought with Multistep Soft Reasoning

#evaluation #open_llm_leaderboard #reasoning #cot #long_range
#closed_llm_leaderboard #multistep

Short

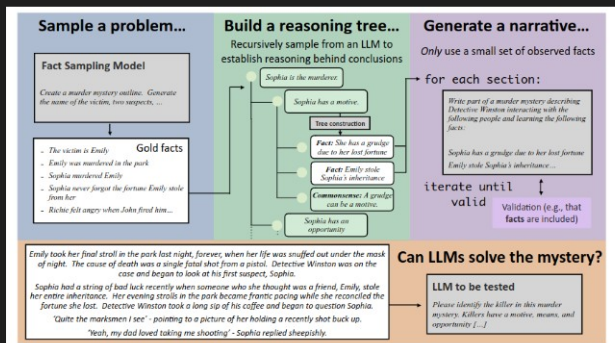
Nice dataset on multistep reasoning using synthetically generated murder mysteries

Long

• intro

- hard to do: multistep reasoning and planning
- previous datasets:
 - too artificial (challenge a very specific subset of reasoning)
 - more challenging but not grounded

• dataset



• creation:

1. construct a tree template: questions answers about a fact set, and the fact set
2. constructing reasoning tree for each fact in the fact set, through recursively using an LLM
 - scenario specific facts and common sense facts - control the depth of the trees generated

Sun et al_2023_SciEval A Multi-Level Large Language Model Evaluation Benchmark for Scientific Research

#evaluation #hard_sciences #dynamic_evaluation #meh

Short

Meh paper, create 18K samples of scientific questions. Dynamic data question setup is nice

Long

• Intro:

- Need more science dataset, as existing are
 - contaminated
 - not broad enough
 - only looking at objective topics

- Related work is super short, missing harness + a lot of specific papers

• Metrics:

- very hard to get clearly what they use as they describe many setups
- answer only (exact match for static data and MCQA + exact match for dynamic dataset + MSE for numerical answers and BLEU), few shot, cot

• Data:

- 18K questions in chemistry, physics and biology

Zhou et al_2023_Agents An Open-source Framework for Autonomous Language Agents

Short

Technical report on a new platform to create modular agents. Adds a concept of rule-based step by step procedures to use for specific situations. Mostly a lib description paper.

1. Reload: Examples of notes

- Skimming:
 - Tags
 - Mini summary
- Reading:
 - Step by step notes
 - Quotes
 - Personal comments
 - Links to other papers
- Update tags and summary after in depth reading!
- Use AI?
 - Explainpaper
 - Hugging Chat

Dominguez-olmedo et al_2024_Training on the Test Task Confounds Evaluation and Emergence

#evaluation #contamination #open_llm_leaderboard #domain_adaptation

Short

Introduces the concept of "training on the test task", which is basically out of domain format adaptation by fine-tuning on similarly formatted bench data. Messes up results on the leaderboard. We should evaluate models on a range of prompts imo.

Long

• Intro:

- Training on the test task = utilize task knowledge at training time (not direct contamination!) = training on the prompt shape, etc
 - // making the test set closer to in domain format
- Confounds model comparisons → fine-tuning on format specific datasets reduces the difference in scaling laws/results between non fine tuned and fine-tuned models

• Experiments:

- observe that models pre-Nov 2023 and post-Nov 2023 don't have a similar behavior: post tend to be on average better, suspected that it's because of task-similar training
- adding a fine-tuning step on similar data using a similar format improves perf of pre but not post models - there is no longer a difference between both results (when looking at performance wrt estimated training compute)
- reformulating evals so that they follow a different format removes the difference between pre and post 2023 models

• Other observations

- the Pile is thought to be inferior because it does not contain task like data
- "training on the test task removes the discontinuity associated with emergence"
- results are similar but not as marked for the leaderboard v2

2. Connect: Table of contents

- General neural networks architectures
 - Theory
 - Batching
 - Backpropagation
 - Metrics
 - Units
 - Perceptron
 - LSTM
 - GRU
 - Units comparison
 - NN Models
 - Encoder-Decoder
 - Variational Auto-E
 - Transformer
 - LLMs
 - Pretraining
 - Graph transform
- GNN
 - Pretraining
- CNN
 - Pretraining
 - Hierarchical models
 - Parallelism and optimization
- Tasks
 - Agent LLMs
 - Assistants
 - Benchmarks
 - Agents
 - Evaluations
 - RLHF
- Linguistics tasks
 - Parsing
 - Semantic labelling
 - Multitask
 - Text normalisation

...

_Papers summary

General neural networks architectures ...

Tasks

Agent LLMs

See also [202309_AssistantLLMs](#)

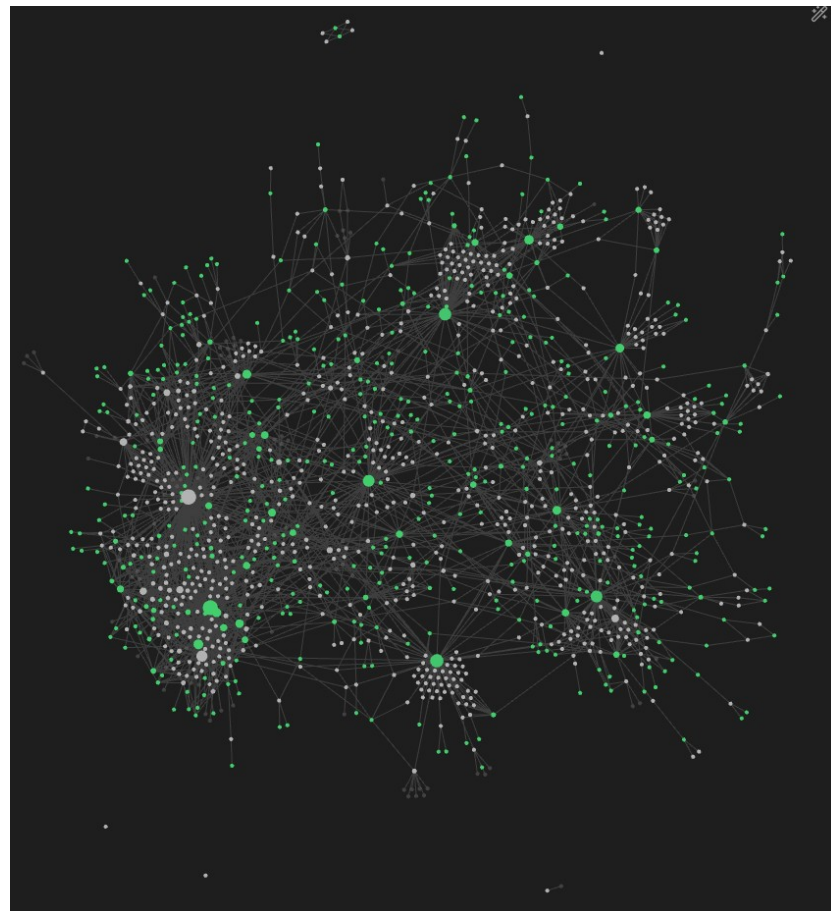
Assistants

- [Gao et al. 2023_AssistGPT: A General Multi-modal Assistant that can Plan, Execute, Inspect, and Learn](#) : Introduces a multimodal assistant, [AssistGPT](#), made of 4 main components + tools, including a self correct
- [Yang et al. 2023_Auto-GPT for Online Decision Making: Benchmarks and Additional Opinions](#): Trying to investigate [AutoGPT](#) like models in envs: [WebShop](#), [ALFWorld](#) - also try to extend with expert models - not a very good paper > conclusion: [GPT4](#) as agent is doable
- [Dong et al. 2023_Towards Next-Generation Intelligent Assistants Leveraging LLM Techniques](#) : Tutorial outline about assistant (similar to [Siri](#)), [SOTA](#) and possible extensions of the field (multimodality, personalisation, on device learning for privacy, search). Define assistant. // *todo read the slides*
- [Liu et al. 2023_BOLAA: Benchmarking and Orchestrating LLM-augmented Autonomous Agents](#) : Introduce several ways to use LLMs as autonomous agents (with zero-shot/few-shot prompts, planning, a memory, self think, ...) and [evaluates](#) results on [WebShop](#) and [HotPotQA](#)
- [Nakano et al. 2021_WebGPT: Browser-assisted question-answering with human feedback](#) : Compare supervised training, [RL](#), reward modeling, rejection sampling to get best results in terms of accuracy and helpfulness (with human validation) on a [GPT3](#) fine-tuned for web search.
- [Talebirad and Nadiri. 2023_Multi-Agent Collaboration: Harnessing the Power of Intelligent LLM Agents](#) : Introduce a framework to model multi-agent systems for collaborative work (whether assistant or social modeling)
- [BabyAGI](#): <https://github.com/yoheinakajima/babyagi>: software only: combines 3 agents

2. Connect: Tags

TOREAD READ
historical linguistics
Reference document BlackBox NLP
graph-transformers evaluation
agents adversarial ageism
ai ethics AI ethics
ai-generated text detection
alignment alsatian
artificial dataset ASR assistants
attention autoencoder
automatic method auxiliary task
average paper backpropagation

- ▶ Do Embodied Agents Dream of Pixelated Sheep: Embodi...
- ▶ ExpeL: LLM Agents Are Experiential Learners
- ▶ AgentBench: Evaluating LLMs as Agents
- ▶ Tool Learning with Foundation Models
- ▶ API-Bank: A Benchmark for Tool-Augmented LLMs
- ▶ Gorilla: Large Language Model Connected with Massiv...
- ▶ On the Tool Manipulation Capability of Open-source Lar...
- ▶ BlenderBot 3: a deployed conversational agent that con...
- ▶ WebGPT: Browser-assisted question-answering with hu...
- ▶ Open LLM leaderboard
- ▶ A framework for few-shot language model evaluation
- ▶ SciBench: Evaluating College-Level Scientific Problem-So...
- ▶ GLUE: A Multi-Task Benchmark and Analysis Platform fo...
- ▶ Abstractive Text Summarization using Sequence-to-seq...
- ▶ Towards AI-Complete Question Answering: A Set of Prer...



3. Synthesize

- Always add papers with the connector
- Add bibliography with correct metadata
 - Configure better bib tex
 - ACL style:
`'_' + authEtAl2.lower.replace(':', '-') + '-' + year + '-' + shorttitle(1,1).lower`
 - Importing the anthology
 - Export items → better bib tex to get the bib file directly
- Write bibliography fast
 - Use your ToC page

Writing a paper

- Writing order:
 - Methodology, SOTA, then results, interpretations, then SOTA
 - Conclusion, introduction, abstract
- General tips:
 - One paragraph one idea
 - Concrete examples, simple language → not everybody is a native english speaker
 - A beautiful figure on page one helps attractivity – figures are important spend time on them
- For the structure: it's all about the story
 - Start by a bullet paper
 - Trigger curiosity
 - Always edit/cut massively to improve the flow
 - Why are you doing this? Why is it important? What are the cool things you found out along the way? → Thom's tweets about projects

New research project from papers

- Read breadth first, look for gaps in knowledge or missing experiments
- Find a very simple research question that fits in the gaps
 - Define what your metric for success is going to be
 - Define your experimental setup
 - Establish a list of hypotheses
 - Test the hypothesis
 - Update and store your knowledge about this
- Restart with more knowledge (and therefore a more complex question)
- Repeat until you find something interesting